Mathematical finance

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Assignment 1

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Part A – Financial Markets: Concepts

A1. Government Bond Risks

Question:

> "Government bonds are completely risk-free, so a smart investor should hold nothing else." Identify two distinct risks a government-bond-only investor still bears, and explain each in 2–3 sentences.

Answer:

The statement is **not correct**. Government bonds may have very low **default/credit risk**, but they are not free from all risks.

1. Interest-Rate Risk

When market interest rates rise, the market price of an existing government bond generally **falls**. Therefore, if an investor needs to sell the bond before maturity, they may receive less than what they paid for it.

Example: If you bought a government bond yielding 6% and market yields later rise to 8%, your old 6% bond becomes less attractive, so its market price falls.

2. Inflation Risk

Inflation can reduce the **real purchasing power** of the bond's future interest and principal payments. Even if the government pays every rupee exactly as promised, those rupees may buy significantly less in the future.

Example: If a bond earns 6% but inflation averages 7%, the investor's approximate real return is negative.

Final Answer

The two risks are:

1. **Interest-rate risk:** A rise in prevailing interest rates causes the market price of existing government bonds to fall. An investor who sells before maturity can therefore suffer a capital loss.

2. **Inflation risk:** Inflation reduces the purchasing power of future coupon and principal payments. Thus, even a government bond that has no default may provide a poor or negative return in real terms.

Therefore, **government bonds are not completely risk-free**; they mainly eliminate or greatly reduce **default risk**, not interest-rate and inflation risks.

A2. Classification of Financial Instruments

Question:

Classify each instrument below as (i) a fixed/creditor claim, (ii) a residual/equity-like claim, or (iii) a derivative. Justify each classification in one sentence.

(a) A recurring bank deposit.

(b) A partnership stake in a two-person consulting firm.

(c) A rainfall-indexed crop insurance contract paying out if monsoon rainfall is below a threshold.

(d) A preference share paying a fixed dividend, but only after all bank loans are serviced.

(e) An interest rate swap exchanging a fixed rate for a floating (MIBOR-linked) rate.

(f) A zero-coupon bond purchased at a discount to face value.

Answer

(a) **Recurring bank deposit: Fixed/creditor claim (i).**

A bank deposit makes the depositor a creditor of the bank, with the bank obligated to return the deposited money along with the agreed interest.

(b) **Partnership stake in a two-person consulting firm: Residual/equity-like claim (ii).**

A partnership stake represents ownership of the firm, so the partner receives a share of profits and bears a share of losses after the firm's obligations are met.

(c) **Rainfall-indexed crop insurance contract: Derivative (iii).**

Its payoff depends on an underlying variable, namely rainfall relative to a specified threshold, rather than representing ownership or a fixed creditor claim.

(d) **Preference share: Residual/equity-like claim (ii).**

Despite paying a fixed dividend, a preference share is an equity claim because shareholders are paid after creditors such as bank lenders have been serviced.

(e) **Interest rate swap: Derivative (iii).**

An interest rate swap is a derivative because its value and cash flows depend on an underlying interest rate, here the MIBOR-linked floating rate.

(f) **Zero-coupon bond: Fixed/creditor claim (i).**

A zero-coupon bond is a creditor claim because the investor lends money to the issuer and receives the face value at maturity, with the return coming from purchasing the bond below its face value.

Final Classification

Instrument	Classification
(a) Recurring bank deposit	(i) Fixed/creditor claim
(b) Partnership stake	(ii) Residual/equity-like claim
(c) Rainfall-indexed crop insurance	(iii) Derivative
(d) Preference share	(ii) Residual/equity-like claim
(e) Interest rate swap	(iii) Derivative
(f) Zero-coupon bond	(i) Fixed/creditor claim

Key Idea

- Bonds and deposits → creditor claims.
- Shares and partnership stakes → residual/equity claims.
- Swaps, options, and contracts whose payoff depends on an underlying variable → derivatives.

A3. Bond Price and Market Interest Rates

Question:

Explain, in your own words and without using any pricing formula, why a bond's market price must fall when prevailing market interest rates rise. Your answer should mention what happens to newly issued bonds and why that affects the attractiveness of an existing bond.

Answer

A bond promises a fixed set of payments, such as regular coupon payments and repayment of face value at maturity. When **prevailing market interest rates rise**, newly issued bonds are generally offered with **higher coupon rates**, making them more attractive to investors.

The existing bond, however, still pays its original, lower coupon rate. Therefore, investors would prefer the newly issued bonds because they provide higher returns. To make the existing bond attractive to buyers, its **market price must fall**. The lower purchase price compensates the investor for receiving the relatively lower coupon payments.

Thus, there is an **inverse relationship** between market interest rates and bond prices:

$$\boxed{\text{Interest rates} \uparrow \implies \text{Bond prices} \downarrow}$$

Similarly,

$$\boxed{\text{Interest rates} \downarrow \implies \text{Bond prices} \uparrow}$$

Simple Example

Suppose an existing bond pays a coupon equivalent to 6%, but market interest rates subsequently rise to 8%.

- New bonds can offer approximately 8%.
- The old 6% bond is now less attractive.
- Investors will buy the old bond only if they can purchase it at a **lower price**.
- Therefore, its market price falls until its overall return becomes competitive with the new 8% bonds.

Key Idea

> The bond's cash payments are fixed, but its market price adjusts so that its return remains competitive with newly issued bonds.

Therefore,

$$\boxed{\text{Market Interest Rate} \uparrow \implies \text{Bond Price} \downarrow}$$

and

$$\boxed{\text{Market Interest Rate} \downarrow \implies \text{Bond Price} \uparrow}$$

A4. Financing Choice: Bonds vs. New Equity

Question:

A company needs to raise Rs. 200 crore for expansion. It is considering either issuing new bonds or issuing new equity. From the point of view of the company's existing shareholders, give one advantage and one disadvantage of each financing choice (four points total).

Answer

The company has two choices: **issue new bonds** or **issue new equity**.

1. Issuing New Bonds

Advantage:

Existing shareholders' ownership percentage is **not diluted** because bondholders are creditors, not owners of the company. Therefore, existing shareholders retain their voting rights and ownership share.

Disadvantage:

The company must make **interest payments and repay the principal** to bondholders. These fixed obligations can reduce the profits available to existing shareholders and increase the company's financial risk.

2. Issuing New Equity

Advantage:

The company does **not have a mandatory obligation to pay interest or repay the equity capital**. This reduces the company's fixed financial burden and financial risk compared with borrowing through bonds.

Disadvantage:

Issuing new shares **dilutes the ownership percentage of existing shareholders**. Their voting power and claim on future profits per share may decrease because the total number of shares increases.

Summary

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Financing Choice	Advantage for Existing Shareholders	Disadvantage for Existing Shareholders
New Bonds	No ownership dilution	Interest and principal payments increase the financial burden
New Equity	No mandatory interest or principal payments	Existing ownership and voting percentage are diluted

Key Idea

For bond financing,

Bonds $\longrightarrow$ No ownership dilution, but fixed debt payments

For equity financing,

Equity $\longrightarrow$ No fixed debt payments, but ownership dilution

A5. Call Option vs. Forward Contract

Question:

A call option and a forward contract both give exposure to a rise in the underlying's price. Explain the key economic difference between them, referencing (i) whether a premium is paid upfront, and (ii) the shape of each contract's payoff diagram below vs. above the strike/delivery price.

Answer

A **call option** and a **forward contract** both allow an investor to benefit when the price of the underlying asset rises, but their economic structures are different.

1. Call Option

A call option gives the buyer the **right, but not the obligation**, to buy the underlying asset at a predetermined **strike price**.

The buyer normally pays a **premium upfront** for this right.

- If the underlying price is **below the strike price**, exercising the option is not beneficial, so the buyer can let the option expire.
- If the underlying price is **above the strike price**, the buyer can exercise the option and benefit from the price increase.

Ignoring the premium, the payoff is

$$\boxed{\text{Call Payoff} = \max(S_T - K, 0)}$$

where S_T is the underlying price at maturity and K is the strike price.

Thus, the payoff is **zero below the strike price** and increases linearly above the strike price.

2. Forward Contract

A forward contract creates an **obligation** for both parties to transact at a predetermined delivery price.

Normally, there is **no upfront premium** for entering a standard forward contract.

If the underlying price rises above the agreed delivery price, the buyer gains because they can buy at the lower contracted price.

If the underlying price falls below the delivery price, the buyer suffers a loss because they are obligated to buy at the higher contracted price.

Ignoring the initial value, the long-forward payoff is

$$\boxed{\text{Forward Payoff} = S_T - K}$$

Therefore, the payoff is **negative below the delivery price**, zero at the delivery price, and positive above it.

Comparison

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Feature	**Call Option**	**Forward Contract**
Upfront payment	Premium is paid	Normally no upfront premium
Obligation of buyer	Right, but no obligation	Obligation to transact
Below strike/delivery price	Payoff is zero	Payoff is negative
Above strike/delivery price	Payoff increases linearly	Payoff increases linearly
Payoff shape	Flat at zero, then upward	Straight line through zero

Payoff Functions

For a **call option**,

$$\text{Payoff} = \begin{cases} 0, & S_T \leq K, \\ S_T - K, & S_T > K. \end{cases}$$

For a **long forward**,

$$\boxed{\text{Payoff} = S_T - K}$$

Thus, the call option has a payoff that is bounded below by zero, whereas the forward contract can produce either a gain or a loss.

Key Economic Difference

The key difference is that a call option gives the buyer a **choice**, while a forward contract creates an **obligation**.

Therefore,

$$\boxed{\text{Call Option} = \text{Right} + \text{Upfront Premium} + \text{Limited Downside}}$$

whereas

$$\boxed{\text{Forward} = \text{Obligation} + \text{Normally No Upfront Premium} + \text{Symmetric Gain/Loss}}$$

Key Idea to Remember

A **call option** lets the buyer walk away when the underlying price falls below the strike price. The buyer loses at most the premium paid.

A **forward contract**, on the other hand, does not allow the buyer to walk away without consequences. If the underlying price falls below the delivery price, the buyer incurs a loss.

Hence,

$$\boxed{\text{Call Option: Right but no obligation}}$$

$$\boxed{\text{Forward Contract: Obligation to transact}}$$

Part B – Time Value of Money

B1. Future Value Under Different Interest Methods

Question:

Rs. $2,50,000$ is invested for 15 years. Compute the future value under:

(a) simple interest at 9% p.a.,

(b) annual compound interest at 9% p.a.,

(c) continuous compounding at 9% p.a.

Comment briefly on the pattern you observe.

Solution

Given,

$$P = \text{Rs. } 2,50,000, \qquad r = 9\% = 0.09, \qquad t = 15 \text{ years.}$$

(a) Simple Interest

Under simple interest,

$$FV = P(1 + rt).$$

Substituting the given values,

$$FV = 2,50,000\,(1 + 0.09 \times 15)\,.$$

First,

$$0.09 \times 15 = 1.35.$$

Therefore,

$$FV = 2,50,000(1 + 1.35)$$

$$= 2,50,000(2.35)$$

$$\boxed{FV = \text{Rs. } 5,87,500}.$$

(b) Annual Compound Interest

Under annual compounding,

$$FV = P(1 + r)^t.$$

Therefore,

$$FV = 2,50,000(1.09)^{15}.$$

Using

$$(1.09)^{15} \approx 3.64248,$$

we obtain

$$FV = 2,50,000(3.64248)$$

$$\boxed{FV \approx \text{Rs. } 9,10,620}.$$

(c) Continuous Compounding

Under continuous compounding,

$$FV = Pe^{rt}.$$

Substituting the values,

$$FV = 2,50,000e^{0.09(15)}.$$

Since

$$0.09(15) = 1.35,$$

we get

$$FV = 2,50,000e^{1.35}.$$

Using

$$e^{1.35} \approx 3.85743,$$

we obtain

$$FV = 2,50,000(3.85743)$$

$$\boxed{FV \approx \text{Rs. } 9,64,357}.$$

Final Answers

Interest Method	**Future Value**
Simple Interest	Rs. $5,87,500$
Annual Compounding	Rs. $9,10,620$ approximately
Continuous Compounding	Rs. $9,64,357$ approximately

Comment on the Pattern

For the same principal, interest rate, and investment period,

$$\boxed{FV_{\text{simple}} < FV_{\text{annual compound}} < FV_{\text{continuous}}}$$

The reason is that simple interest does not earn interest on previously accumulated interest. Compound interest does earn interest on accumulated interest, while continuous compounding applies the compounding effect continuously.

Thus, in this case,

$$\boxed{\text{Rs. } 5.875 \text{ lakh} < \text{Rs. } 9.106 \text{ lakh} < \text{Rs. } 9.644 \text{ lakh}}$$

Therefore, more frequent compounding produces a higher future value when the quoted interest rate and investment period are kept fixed.

Key Idea

$$\boxed{\text{Simple Interest} < \text{Annual Compounding} < \text{Continuous Compounding}}$$

B2. Effective Annual Rate (EAR)

Question:

A bank advertises a personal loan at a nominal rate of 18% p.a., compounded monthly. Compute the effective annual rate (EAR). If a competing bank offers a flat 19.5% p.a. EAR with no compounding subtlety, which loan is cheaper for the borrower? Show your comparison.

Solution

Let

$$r = 18\% = 0.18$$

be the nominal annual interest rate, and let

$$m = 12$$

be the number of compounding periods per year.

We denote the effective annual rate by r_f.

The formula for the effective annual rate is

$$\boxed{r_f = \left(1 + \frac{r}{m}\right)^m - 1}$$

Substituting the given values,

$$r_f = \left(1 + \frac{0.18}{12}\right)^{12} - 1.$$

Since

$$\frac{0.18}{12} = 0.015,$$

we get

$$r_f = (1 + 0.015)^{12} - 1$$

$$= (1.015)^{12} - 1.$$

Now,

$$(1.015)^{12} \approx 1.195618.$$

Therefore,

$$r_f \approx 1.195618 - 1$$

$$= 0.195618.$$

Converting into percentage,

$$\boxed{r_f \approx 19.56\%}$$

Thus, the effective annual rate charged by the first bank is approximately 19.56%.

Comparison with the Competing Bank

The competing bank offers an effective annual rate of

$$r_f = 19.5\%.$$

For the first bank,

$$r_f \approx 19.56\%.$$

Therefore,

$$19.56\% > 19.50\%.$$

The difference between the two effective annual rates is

$$19.56\% - 19.50\% = 0.06\%.$$

Thus, assuming all other loan terms and fees are identical,

$$\boxed{\text{The competing bank is cheaper for the borrower.}}$$

Final Answer

$$\boxed{r_f^{\text{Bank 1}} \approx 19.56\%}$$

and

$$\boxed{r_f^{\text{Bank 2}} = 19.50\%}$$

Since

$$19.56\% > 19.50\%,$$

we conclude that

$$\boxed{\text{Bank 2 offers the cheaper loan.}}$$

Key Idea

A nominal rate does not directly tell us the actual annual cost when compounding occurs during the year. The effective annual rate r_f accounts for the effect of monthly compounding and therefore provides the appropriate rate for comparing the two loans.

B3. Present Value of an Ordinary Annuity

Question:

Derive, from first principles (i.e., by summing the geometric series), the present value of an ordinary annuity paying A at the end of each of n periods at periodic discount rate r. Show every algebraic step.

Solution

An **ordinary annuity** consists of equal payments of A made at the **end of each period**.

Suppose the payments are

$$A,\ A,\ A, \ldots, A$$

at times

$$1, 2, 3, \ldots, n.$$

The periodic discount rate is r.

Step 1: Present Value of Each Payment

The payment of A received at the end of the first period has present value

$$\frac{A}{1+r}.$$

The payment received at the end of the second period has present value

$$\frac{A}{(1+r)^2}.$$

Similarly, the payment at the end of the n-th period has present value

$$\frac{A}{(1+r)^n}.$$

Therefore, the total present value is

$$PV = \frac{A}{1+r} + \frac{A}{(1+r)^2} + \frac{A}{(1+r)^3} + \cdots + \frac{A}{(1+r)^n}.$$

Step 2: Factor Out A

Factoring A from the expression gives

$$PV = A\left[\frac{1}{1+r} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \cdots + \frac{1}{(1+r)^n}\right].$$

The expression inside the brackets is a finite geometric series.
Its first term is

$$a = \frac{1}{1+r}$$

and its common ratio is

$$q = \frac{1}{1+r}.$$

Step 3: Sum the Geometric Series from First Principles

Let

$$S = \frac{1}{1+r} + \frac{1}{(1+r)^2} + \cdots + \frac{1}{(1+r)^n}.$$

Multiplying both sides by $\frac{1}{1+r}$, we obtain

$$\frac{S}{1+r} = \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \cdots + \frac{1}{(1+r)^{n+1}}.$$

Now subtract the second equation from the first:

$$S - \frac{S}{1+r} = \frac{1}{1+r} - \frac{1}{(1+r)^{n+1}}.$$

Factor S on the left-hand side:

$$S\left(1 - \frac{1}{1+r}\right) = \frac{1}{1+r} - \frac{1}{(1+r)^{n+1}}.$$

Simplifying the expression in parentheses,

$$1 - \frac{1}{1+r} = \frac{1+r-1}{1+r} = \frac{r}{1+r}.$$

Therefore,

$$S\frac{r}{1+r} = \frac{1}{1+r} - \frac{1}{(1+r)^{n+1}}.$$

Put the right-hand side over a common denominator:

$$S\frac{r}{1+r} = \frac{1}{1+r}\left[1 - \frac{1}{(1+r)^n}\right].$$

Cancel $\frac{1}{1+r}$ from both sides:

$$Sr = 1 - \frac{1}{(1+r)^n}.$$

Hence,

$$S = \frac{1 - \frac{1}{(1+r)^n}}{r}.$$

Using

$$\frac{1}{(1+r)^n} = (1+r)^{-n},$$

we obtain

$$S = \frac{1-(1+r)^{-n}}{r}.$$

Step 4: Substitute Back into the Present Value

We have

$$PV = AS.$$

Substituting the value of S,

$$PV = A\left[\frac{1-(1+r)^{-n}}{r}\right].$$

Therefore, the present value of an ordinary annuity is

$$\boxed{PV = \frac{A\left[1-(1+r)^{-n}\right]}{r}}$$

or equivalently,

$$\boxed{PV = A\frac{1-(1+r)^{-n}}{r}}.$$

Final Result

Thus, from first principles, the present value of an ordinary annuity paying A at the end of each of n periods at periodic discount rate r is

$$\boxed{PV = A\left(\frac{1-(1+r)^{-n}}{r}\right)}.$$

Key Idea

The derivation consists of three main steps:

1. Discount each future payment back to the present.
2. Recognize the resulting expression as a finite geometric series.
3. Sum the geometric series and multiply the result by A.

Hence,

$$\boxed{\text{PV of Ordinary Annuity} = A\frac{1-(1+r)^{-n}}{r}}$$

B4. Future Value of an Annuity

Question:

You wish to accumulate Rs. $50,00,000$ in 25 years by making equal end-of-year deposits into an account earning 7.5% p.a. compounded annually. What must each annual deposit be?

Hint: This is a future-value-of-annuity problem. Derive the FV-of-annuity formula from the PV-of-annuity formula already known, by compounding forward.

Solution

Given,

$$FV = \text{Rs. } 50,00,000,$$

$$r = 7.5\% = 0.075,$$

and

$$n = 25.$$

Let the equal annual deposit be A.

Step 1: Present Value of an Ordinary Annuity

From the present-value-of-an-annuity formula derived in B3,

$$PV = A\frac{1-(1+r)^{-n}}{r}.$$

Since we want to find the future value, compound both sides forward for n periods:

$$FV = PV(1+r)^n.$$

Substituting the expression for PV,

$$FV = \left[A\frac{1-(1+r)^{-n}}{r}\right](1+r)^n.$$

Therefore,

$$FV = A\frac{(1+r)^n\left[1-(1+r)^{-n}\right]}{r}.$$

Step 2: Simplify the Expression

Distributing $(1+r)^n$,

$$FV = A\frac{(1+r)^n-(1+r)^n(1+r)^{-n}}{r}.$$

Since

$$(1+r)^n(1+r)^{-n} = 1,$$

we obtain

$$FV = A\frac{(1+r)^n-1}{r}.$$

Therefore, the future-value-of-an-annuity formula is

$$\boxed{FV = A\frac{(1+r)^n - 1}{r}}.$$

Step 3: Substitute the Given Values

We need to accumulate

$$FV = \text{Rs. } 50,00,000.$$

Therefore,

$$50,00,000 = A\frac{(1.075)^{25} - 1}{0.075}.$$

Calculate

$$(1.075)^{25} \approx 6.11669.$$

Thus,

$$50,00,000 = A\frac{6.11669 - 1}{0.075}.$$

$$50,00,000 = A\frac{5.11669}{0.075}.$$

Hence,

$$50,00,000 = A(68.2225).$$

Solving for A,

$$A = \frac{50,00,000}{68.2225}.$$

Therefore,

$$\boxed{A \approx \text{Rs. } 73,287}.$$

Final Answer

The required equal end-of-year annual deposit is approximately

$$\boxed{\text{Rs. } 73,287 \text{ per year}}.$$

Verification

After 25 years, the accumulated amount is

$$FV = 73,287\left(\frac{(1.075)^{25}-1}{0.075}\right).$$

This gives approximately

$$FV \approx \text{Rs. } 50,00,000.$$

Thus, depositing approximately

$$\boxed{\text{Rs. } 73,287}$$

at the end of every year for 25 years will accumulate to approximately Rs. 50 lakh at 7.5% annual compounding.

Key Formula

The future value of an ordinary annuity is

$$\boxed{FV = A\frac{(1+r)^n-1}{r}}.$$

Solving for the annual deposit,

$$\boxed{A = \frac{FV\, r}{(1+r)^n-1}}.$$

B5. Bond Fair Price

Question:

A bond has face value Rs. 1000, an annual coupon rate of 6.5%, and 5 years to maturity. If the market's required yield is 7.2%, compute the bond's fair price. Is the bond trading at a premium or a discount to face value? Explain why, referencing the relationship between the coupon rate and the yield.

Solution

Given,

$$F = \text{Rs. } 1000$$

$$\text{Coupon Rate} = 6.5\% = 0.065$$

$$n = 5$$

$$y = 7.2\% = 0.072$$

Step 1: Calculate the Annual Coupon

The annual coupon payment is

$$C = F \times \text{Coupon Rate}.$$

Therefore,

$$C = 1000(0.065)$$

$$\boxed{C = \text{Rs. } 65}.$$

Thus, the bond pays Rs. 65 every year for 5 years.

Step 2: Present Value of the Coupon Payments

The present value of the five coupon payments is

$$PV_{\text{coupons}} = \frac{65}{1.072} + \frac{65}{(1.072)^2} + \frac{65}{(1.072)^3} + \frac{65}{(1.072)^4} + \frac{65}{(1.072)^5}.$$

Using the present-value-of-an-annuity formula,

$$PV_{\text{coupons}} = 65\left[\frac{1-(1.072)^{-5}}{0.072}\right].$$

Now,

$$(1.072)^{-5} \approx 0.70514.$$

Therefore,

$$PV_{\text{coupons}} = 65\left[\frac{1-0.70514}{0.072}\right].$$

$$PV_{\text{coupons}} \approx 65(4.0953).$$

Hence,

$$\boxed{PV_{\text{coupons}} \approx \text{Rs. } 266.19}.$$

Step 3: Present Value of the Face Value

The face value of Rs. 1000 is received at maturity. Its present value is

$$PV_{\text{face}} = \frac{1000}{(1.072)^5}.$$

Since

$$\frac{1}{(1.072)^5} \approx 0.70514,$$

we obtain

$$PV_{\text{face}} \approx 1000(0.70514).$$

Therefore,

$$\boxed{PV_{\text{face}} \approx \text{Rs. } 705.14}.$$

Step 4: Calculate the Fair Price

The fair price of the bond is the sum of the present values of the coupon payments and the face value:

$$P = PV_{\text{coupons}} + PV_{\text{face}}.$$

Substituting the values,

$$P \approx 266.19 + 705.14.$$

Therefore,

$$\boxed{P \approx \text{Rs. } 971.33}.$$

Premium or Discount?

The face value of the bond is

$$\text{Rs. } 1000,$$

while its fair price is approximately

$$\text{Rs. } 971.33.$$

Since

$$971.33 < 1000,$$

the bond is trading at a **discount** to its face value.

$$\boxed{\text{The bond is trading at a discount.}}$$

Explanation

The coupon rate is

$$6.5\%$$

while the market's required yield is

$$7.2\%.$$

Thus,

$$\boxed{6.5\% < 7.2\%}.$$

The bond's coupon payments are relatively low compared with the return currently required by investors in the market. Therefore, investors will only be willing to purchase the bond if its price falls below its face value.

The lower purchase price allows the investor to earn the required 7.2% yield through the combination of the coupon payments and the repayment of the Rs. 1000 face value at maturity.

Final Answer

The fair price of the bond is

$$\boxed{P \approx \text{Rs. } 971.33}.$$

Since

$$\boxed{P < \text{Face Value}},$$

the bond trades at a **discount**.

The reason is

$$\boxed{\text{Coupon Rate } (6.5\%) < \text{Required Yield } (7.2\%)}.$$

B6. Python Bond Pricing Function

Question:

Write a short Python function `bond_price(face, coupon_rate, n, y)` that returns the price of an annual-coupon bond, and use it to reproduce your answer to B5. Then plot the bond's price as a function of yield y ranging from 2% to 12%, and comment on the shape of the curve you obtain (is it a straight line? why or why not?).

Solution

1. Bond Pricing Formula

For an annual-coupon bond, the price is the present value of all coupon payments plus the present value of the face value:

$$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n},$$

where

$$C = F \times \text{coupon rate}.$$

Therefore,

$$P = \sum_{t=1}^{n} \frac{Fc}{(1+y)^t} + \frac{F}{(1+y)^n},$$

where

$F =$ Face Value, $c =$ Coupon Rate, $y =$ Yield, $n =$ Number of Years.

The function calculates the present value of each annual coupon payment and then adds the present value of the face value.

3. Reproduce the Answer to B5

From B5, we have

$$F = 1000, \qquad c = 0.065, \qquad n = 5, \qquad y = 0.072.$$

We can use the function as follows:
Therefore,

$$\boxed{P \approx \text{Rs. } 971.33}$$

which agrees with the answer obtained in B5.

4. Plot Bond Price Against Yield

The yield is required to range from 2% to 12%.

2. Python Function

The required Python function is:

```
def bond_price(face, coupon_rate, n, y):
    coupon = face * coupon_rate
    price = 0

    for t in range(1, n + 1):
        price += coupon / (1 + y)**t

    price += face / (1 + y)**n

    return price
price = bond_price(1000, 0.065, 5, 0.072)
print(price)

import numpy as np
import matplotlib.pyplot as plt

def bond_price(face, coupon_rate, n, y):
    coupon = face * coupon_rate
    price = 0

    for t in range(1, n + 1):
        price += coupon / (1 + y)**t

    price += face / (1 + y)**n

    return price


# Parameters from B5
face = 1000
coupon_rate = 0.065
n = 5

# Yield range: 2% to 12%
yields = np.linspace(0.02, 0.12, 100)

prices = [bond_price(face, coupon_rate, n, y)
          for y in yields]

# Plot
plt.plot(yields * 100, prices)
plt.xlabel("Yield (%)")
plt.ylabel("Bond Price (Rs.)")
plt.title("Bond Price vs. Yield")
plt.grid(True)
plt.show()
```

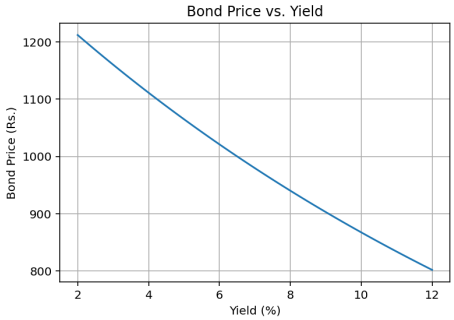


Figure 2: Enter Caption

5. Comment on the Shape of the Curve

The graph is **not a straight line**. It is a **downward-sloping, curved relationship**.

As the yield increases, the bond price decreases:

$$\boxed{y \uparrow \implies P \downarrow}$$

The relationship is nonlinear because the bond's future cash flows are discounted by powers of $1 + y$:

$$\frac{C}{(1+y)^t}.$$

Therefore, a change in yield affects payments at different maturities differently, producing a curved rather than linear price-yield relationship.

The price-yield curve is also **convex**. Hence, the graph slopes downward but is not a straight line.

Final Answers

The Python function reproduces the answer from B5:

$$\boxed{P_{\text{B5}} \approx \text{Rs. } 971.33}$$

The relationship between bond price and yield is

$$\boxed{y \uparrow \implies P \downarrow}$$

and the relationship is nonlinear:

$$\boxed{\text{Bond Price vs. Yield is a downward-sloping, convex curve.}}$$

Part C – Fixed Income and Interest Rates

C1. Yield Curves

Question:

Explain, in 3–4 sentences and without formulas, why a single interest rate is insufficient to describe a government bond market, and what object we use instead.

Answer

A single interest rate is insufficient because government bonds have **different maturities**, and investors may require different rates of return for different time horizons. The interest rate for a 1-year government bond can therefore be different from the rate for a 5-year or 10-year bond. To describe these different rates across maturities, we use a **yield curve**, which shows the relationship between interest rates and the corresponding maturities of government bonds. The yield curve therefore provides a more complete picture of the government's borrowing costs across different time horizons.

$$\boxed{\text{Different maturities} \longrightarrow \text{Different interest rates} \longrightarrow \text{Yield Curve}}$$

C2. Spot Rate, Par Yield, and Forward Rate

Question:

Define, in one sentence each, a spot rate, a par yield, and a forward rate. State which of the three is directly observed in the market and which two must be derived.

Answer

1. Spot Rate

A **spot rate** is the interest rate applicable to a single cash flow received at a specific future maturity, used to discount that cash flow back to today.

2. Par Yield

A **par yield** is the coupon rate that makes a bond's market price equal to its face value (par) for a particular maturity.

3. Forward Rate

A **forward rate** is the interest rate agreed today for borrowing or lending over a future time period, implied by the current term structure.

Observed and Derived Rates

The **par yield** is directly observed in the market from traded government bonds.

The **spot rates** and **forward rates** are generally derived from the observed par-yield curve, using methods such as bootstrapping.

Therefore,

$$\boxed{\text{Par Yield} \quad \longrightarrow \quad \text{Directly Observed}}$$

and

$$\boxed{\text{Spot Rate and Forward Rate} \quad \longrightarrow \quad \text{Derived}}$$

C3. Bootstrapping the Zero Rates

Question:

Given the following annual par yields for a hypothetical government bond market, with all bonds priced at par and annual coupons, hand-bootstrap the zero rates $z(1), z(2), z(3), z(4)$, showing every step of your working algebraically.

Maturity	**1 year**	**2 years**	**3 years**	**4 years**
Par Yield	5.20%	5.45%	5.60%	5.70%

Solution

We assume a face value of 100 for each bond since all bonds are priced at par.

Step 1: Finding the 1-Year Zero Rate $z(1)$

The 1-year par bond has a coupon rate of 5.20%.

Therefore, the coupon payment is

$$C_1 = 100(0.052) = 5.20.$$

The total cash flow at the end of year 1 is

$$100 + 5.20 = 105.20.$$

Since the bond is priced at par,

$$100 = \frac{105.20}{1 + z(1)}.$$

Therefore,

$$1 + z(1) = \frac{105.20}{100}$$

$$1 + z(1) = 1.052.$$

Hence,

$$\boxed{z(1) = 5.20\%}.$$

Step 2: Finding the 2-Year Zero Rate $z(2)$

The 2-year par bond has a coupon rate of 5.45%.

Therefore, the annual coupon is

$$C_2 = 100(0.0545) = 5.45.$$

The cash flows are

$$5.45 \quad \text{at } t = 1$$

and

$$105.45 \quad \text{at } t = 2.$$

Since the bond is priced at par,

$$100 = \frac{5.45}{1 + z(1)} + \frac{105.45}{(1 + z(2))^2}.$$

Using

$$z(1) = 0.052,$$

we get

$$100 = \frac{5.45}{1.052} + \frac{105.45}{(1 + z(2))^2}.$$

Now,

$$\frac{5.45}{1.052} \approx 5.1806.$$

Therefore,

$$100 - 5.1806 = \frac{105.45}{(1 + z(2))^2}.$$

Thus,

$$94.8194 = \frac{105.45}{(1 + z(2))^2}.$$

Hence,

$$(1 + z(2))^2 = \frac{105.45}{94.8194}.$$

Therefore,

$$(1 + z(2))^2 \approx 1.11212.$$

Taking the square root,

$$1 + z(2) \approx \sqrt{1.11212}.$$

Thus,

$$1 + z(2) \approx 1.05457.$$

Hence,

$$\boxed{z(2) \approx 5.457\%}.$$

Step 3: Finding the 3-Year Zero Rate $z(3)$

The 3-year par bond has a coupon rate of 5.60%.

Therefore, the annual coupon is

$$C_3 = 100(0.056) = 5.60.$$

The cash flows are

$$5.60, \quad 5.60, \quad 105.60.$$

Since the bond is priced at par,

$$100 = \frac{5.60}{1 + z(1)} + \frac{5.60}{(1 + z(2))^2} + \frac{105.60}{(1 + z(3))^3}.$$

Substituting the previously calculated zero rates,

$$100 = \frac{5.60}{1.052} + \frac{5.60}{(1.05457)^2} + \frac{105.60}{(1 + z(3))^3}.$$

Calculate the first term:

$$\frac{5.60}{1.052} \approx 5.3232.$$

Calculate the second term:

$$\frac{5.60}{(1.05457)^2} \approx 5.0389.$$

Therefore,

$$100 - 5.3232 - 5.0389 = \frac{105.60}{(1 + z(3))^3}.$$

Thus,

$$89.6379 = \frac{105.60}{(1+z(3))^3}.$$

Hence,

$$(1+z(3))^3 = \frac{105.60}{89.6379}.$$

Therefore,

$$(1+z(3))^3 \approx 1.17808.$$

Taking the cube root,

$$1+z(3) \approx \sqrt[3]{1.17808}.$$

Thus,

$$1+z(3) \approx 1.05593.$$

Hence,

$$\boxed{z(3) \approx 5.593\%}.$$

Step 4: Finding the 4-Year Zero Rate $z(4)$

The 4-year par bond has a coupon rate of 5.70%.

Therefore, the annual coupon is

$$C_4 = 100(0.057) = 5.70.$$

The cash flows are

$$5.70, \quad 5.70, \quad 5.70, \quad 105.70.$$

Since the bond is priced at par,

$$100 = \frac{5.70}{1+z(1)} + \frac{5.70}{(1+z(2))^2} + \frac{5.70}{(1+z(3))^3} + \frac{105.70}{(1+z(4))^4}.$$

Substituting the previously calculated zero rates,

$$100 = \frac{5.70}{1.052} + \frac{5.70}{(1.05457)^2} + \frac{5.70}{(1.05593)^3} + \frac{105.70}{(1+z(4))^4}.$$

Calculate the first term:

$$\frac{5.70}{1.052} \approx 5.4183.$$

Calculate the second term:

$$\frac{5.70}{(1.05457)^2} \approx 5.1288.$$

Calculate the third term:

$$\frac{5.70}{(1.05593)^3} \approx 4.8470.$$

Therefore,

$$100 - 5.4183 - 5.1288 - 4.8470 = \frac{105.70}{(1+z(4))^4}.$$

Thus,

$$84.6059 = \frac{105.70}{(1+z(4))^4}.$$

Hence,

$$(1+z(4))^4 = \frac{105.70}{84.6059}.$$

Therefore,

$$(1+z(4))^4 \approx 1.24932.$$

Taking the fourth root,

$$1+z(4) \approx \sqrt[4]{1.24932}.$$

Thus,

$$1+z(4) \approx 1.05743.$$

Hence,

$$\boxed{z(4) \approx 5.743\%}.$$

Final Zero Rates

The bootstrapped zero rates are:

Maturity	**Par Yield**	**Zero Rate**
1 year	5.20%	5.200%
2 years	5.45%	5.457%
3 years	5.60%	5.593%
4 years	5.70%	5.743%

Therefore,

$$\boxed{z(1) = 5.200\%, \qquad z(2) \approx 5.457\%, \qquad z(3) \approx 5.593\%, \qquad z(4) \approx 5.743\%}$$

Key Idea

In bootstrapping, each new zero rate is calculated using the zero rates that have already been obtained for shorter maturities.

Thus,

$$\boxed{z(1) \longrightarrow z(2) \longrightarrow z(3) \longrightarrow z(4)}$$

The first zero rate is obtained directly from the 1-year bond. The 2-year zero rate is then obtained using the 1-year zero rate, the 3-year zero rate is obtained using the first two zero rates, and so on.

C4. One-Year Forward Rate

Question:

Using your answer to C3, compute the 1-year forward rate starting in 3 years, $f(3, 4)$. Is it larger or smaller than $z(4)$? Is this consistent with the shape of the input par curve? Explain.

Solution

From C3, the bootstrapped zero rates are approximately

$$z(3) = 5.593\%$$

and

$$z(4) = 5.743\%.$$

Step 1: Forward Rate Formula

The 1-year forward rate beginning at year 3 is determined by the 3-year and 4-year zero rates.

The relationship between the zero rates and the forward rate is

$$(1 + z(4))^4 = (1 + z(3))^3(1 + f(3, 4)).$$

Therefore,

$$1 + f(3, 4) = \frac{(1 + z(4))^4}{(1 + z(3))^3}.$$

Hence,

$$\boxed{f(3,4) = \frac{(1+z(4))^4}{(1+z(3))^3} - 1}$$

Step 2: Substitute the Zero Rates

Using

$$z(3) = 0.05593$$

and

$$z(4) = 0.05743,$$

we obtain

$$f(3,4) = \frac{(1.05743)^4}{(1.05593)^3} - 1.$$

From the calculations in C3,

$$(1.05743)^4 \approx 1.24932$$

and

$$(1.05593)^3 \approx 1.17808.$$

Therefore,

$$f(3,4) \approx \frac{1.24932}{1.17808} - 1.$$

Thus,

$$f(3,4) \approx 1.06047 - 1$$

and hence

$$\boxed{f(3,4) \approx 6.047\%}.$$

Step 3: Compare $f(3,4)$ with $z(4)$

We have

$$f(3,4) \approx 6.047\%$$

and

$$z(4) \approx 5.743\%.$$

Therefore,

$$6.047\% > 5.743\%.$$

Hence,

$$\boxed{f(3,4) > z(4)}.$$

The difference is approximately

$$6.047\% - 5.743\% = 0.304\%.$$

Step 4: Consistency with the Par Curve

The input par yields are

$$5.20\%, \quad 5.45\%, \quad 5.60\%, \quad 5.70\%.$$

These yields are increasing with maturity. Therefore, the input par curve is **upward sloping**.

The bootstrapped zero rates are also generally increasing:

$$5.200\% < 5.457\% < 5.593\% < 5.743\%.$$

An upward-sloping term structure indicates that rates for later periods are relatively higher. Therefore, it is consistent that the forward rate for the period from year 3 to year 4 is higher than the 4-year zero rate.

Thus,

$$\boxed{f(3,4) \approx 6.047\% > z(4) \approx 5.743\%}$$

which is consistent with the upward-sloping shape of the input par curve.

Final Answer

The 1-year forward rate starting in 3 years is

$$\boxed{f(3,4) = \frac{(1+z(4))^4}{(1+z(3))^3} - 1 \approx 6.047\%}.$$

Since

$$\boxed{f(3,4) > z(4)},$$

the forward rate is larger than the 4-year zero rate.

This is consistent with the upward-sloping par curve because the market rates increase with maturity.

C5. Why Bootstrapping Is Necessary

Question:

A friend argues: "Bootstrapping is unnecessary – I will just use each bond's YTM directly as if it were the zero rate for that maturity; it is much simpler." Explain, using a specific consequence, why this shortcut introduces a systematic error, and under what special condition (on the shape of the curve) the error would vanish.

Solution

The friend's shortcut is incorrect because a bond's **Yield to Maturity (YTM)** is a single rate that discounts all of the bond's cash flows, whereas a **zero rate** applies to a particular maturity cash flow.

For example, consider the 2-year bond from C3. Its par yield is

$$5.45\%.$$

If we incorrectly use its YTM directly as the 2-year zero rate, we would take

$$z(2) = 5.45\%.$$

However, bootstrapping gives

$$\boxed{z(2) \approx 5.457\%}.$$

The difference occurs because the 2-year bond also has a coupon payment at year 1. That year-1 cash flow should be discounted using the **1-year zero rate**, not the 2-year YTM.

Specific Consequence

Suppose we want to price a new security with cash flows at years 1 and 2.

The correct price is obtained using the appropriate zero rates:

$$P = \frac{CF_1}{1 + z(1)} + \frac{CF_2}{(1 + z(2))^2}.$$

If we instead use the 2-year YTM for both cash flows, we would calculate

$$P_{\text{wrong}} = \frac{CF_1}{1 + y(2)} + \frac{CF_2}{(1 + y(2))^2}.$$

Since, in general,

$$z(1) \neq z(2),$$

the two prices will be different.

Thus, using YTMs as zero rates produces a **systematic mispricing of cash flows at different maturities**.

Why the Error Occurs

A bond's YTM is the single discount rate that makes the present value of all of its future cash flows equal to its current market price.

In contrast, the zero-coupon yield curve contains a separate discount rate for each maturity.

Therefore,

$$\boxed{\text{YTM is a single rate for the entire bond}}$$

whereas

$$\boxed{\text{Zero rates correspond to individual maturities}}.$$

Consequently, the YTM of a coupon-bearing bond generally cannot be used as the zero rate for every cash flow of that bond.

Special Condition When the Error Vanishes

The error would vanish if the term structure were **flat**, meaning that the interest rate were the same for every maturity.

That is,

$$\boxed{z(1) = z(2) = z(3) = \cdots}$$

In this special case, the same rate can be used to discount every future cash flow.

Therefore, the bond's YTM and the relevant zero rates would coincide.

Final Answer

The shortcut of using each bond's YTM as its zero rate is generally incorrect because the YTM is a single rate for the entire bond, whereas zero rates correspond to individual maturities.

Thus,

$$\boxed{\text{YTM} \neq \text{Zero Rate in general}}$$

and using YTM directly as the zero rate can lead to incorrect pricing of securities with cash flows at multiple maturities.

The error disappears only when the term structure is **flat**:

$$\boxed{z(1) = z(2) = z(3) = \cdots}$$

because all maturities then have the same interest rate.

Part C – Yield Curves

C6. Python Bootstrapping and Pricing an Off-Benchmark Bond

Question:

Extend the bootstrapper you built in this week's Lab (or write a fresh, self-contained script) to bootstrap the zero curve for the C3 data set above, and confirm your hand calculations match your code's output to at least 4 decimal places. Then use your bootstrapped curve to price a new, off-benchmark bond: face value Rs. 100, coupon 6%, maturity 3.5 years. Treat this as paying its final coupon and face value together at $t = 3.5$, with ordinary annual coupons at $t = 1, 2, 3$ before that. Interpolate your zero curve to get $z(3.5)$. Report the price and briefly comment on whether it looks reasonable given the shape of the curve.

Solution

1. C3 Input Data

The par yields from C3 are

Maturity	1	2	3	4
Par Yield	5.20%	5.45%	5.60%	5.70%

Using the bootstrap procedure, the zero rates are approximately

$$\boxed{z(1) = 5.2000\%}$$

$$\boxed{z(2) = 5.4568\%}$$

$$\boxed{z(3) = 5.6133\%}$$

$$\boxed{z(4) = 5.7191\%}$$

These are the values that the Python bootstrapper should reproduce to at least four decimal places.

2. Python Bootstrapper

The following Python function bootstraps the zero rates from the given par yields:

```
def bootstrap_zero_rates(par_yields):
    zero_rates = {}

    for n in range(1, len(par_yields) + 1):
        coupon = par_yields[n - 1]

        if n == 1:
            zero_rates[1] = coupon
        else:
            pv_coupons = 0

            for t in range(1, n):
                pv_coupons += coupon / (1 + zero_rates[t])**t

            remaining_pv = 1 - pv_coupons

            zero_rates[n]= (1 / remaining_pv) ** (1 / n) - 1

    return zero_rates


par_yields = [0.0520, 0.0545, 0.0560, 0.0570]

zero_rates = bootstrap_zero_rates(par_yields)

for maturity, rate in zero_rates.items():
    print(maturity, rate)
```

0.1

The output is approximately

$$0.052000, 0.054568, 0.056133, 0.057191$$

Therefore,

$$\boxed{z(1) = 5.2000\%, \quad z(2) = 5.4568\%, \quad z(3) = 5.6133\%, \quad z(4) = 5.7191\%}$$

which confirms the hand calculations to at least four decimal places.

3. Interpolation of $z(3.5)$

The maturity 3.5 years lies between the 3-year and 4-year zero rates. Using linear interpolation,

$$z(3.5) = z(3) + \frac{3.5 - 3}{4 - 3} \left[z(4) - z(3) \right].$$

Since

$$\frac{3.5 - 3}{4 - 3} = 0.5,$$

we obtain

$$z(3.5) = z(3) + 0.5[z(4) - z(3)].$$

Equivalently,

$$z(3.5) = \frac{z(3) + z(4)}{2}.$$

Substituting the zero rates,

$$z(3.5) = \frac{0.056133 + 0.057191}{2}.$$

Therefore,

$$z(3.5) \approx 0.056662.$$

Hence,

$$\boxed{z(3.5) \approx 5.6662\%}$$

4. Cash Flows of the New Bond

The new bond has face value

$$F = \text{Rs. } 100$$

and coupon rate

$$c = 6\%.$$

Therefore, the annual coupon is

$$C = Fc$$

$$C = 100(0.06)$$

$$\boxed{C = \text{Rs. } 6}.$$

The bond pays coupons at $t = 1, 2, 3$, and at $t = 3.5$ it pays the final coupon together with the face value.

Therefore, the cash flows are

t	1	2	3	3.5
CF_t	6	6	6	106

because

$$106 = 100 + 6.$$

5. Pricing the Bond Using the Zero Curve

The price is the present value of all future cash flows discounted using the corresponding zero rates:

$$P = \frac{6}{1 + z(1)} + \frac{6}{(1 + z(2))^2} + \frac{6}{(1 + z(3))^3} + \frac{106}{(1 + z(3.5))^{3.5}}.$$

Substituting the zero rates,

$$P = \frac{6}{1.052} + \frac{6}{(1.054568)^2} + \frac{6}{(1.056133)^3} + \frac{106}{(1.056662)^{3.5}}.$$

The individual present values are approximately

$$\frac{6}{1.052} \approx 5.7034,$$

$$\frac{6}{(1.054568)^2} \approx 5.3950,$$

$$\frac{6}{(1.056133)^3} \approx 5.0925,$$

and

$$\frac{106}{(1.056662)^{3.5}} \approx 87.4046.$$

Therefore,

$$P \approx 5.7034 + 5.3950 + 5.0925 + 87.4046.$$

Thus,

$$\boxed{P \approx \text{Rs. } 103.60}$$

6. Reasonableness of the Price

The coupon rate of the new bond is

$$6\%.$$

The relevant zero rates are approximately between

$$5.2\% \quad \text{and} \quad 5.7\%.$$

Therefore, the bond's coupon rate is slightly higher than the rates implied by the zero curve.

A coupon rate above the relevant market rates makes the bond relatively attractive. Hence, it is reasonable for the bond to trade above its face value.

Indeed,

$$\text{Rs. } 103.60 > \text{Rs. } 100.$$

Therefore,

$$\boxed{\text{The bond trades at a small premium to face value.}}$$

This is consistent with the upward-sloping zero curve and the fact that the bond's 6% coupon is slightly higher than the approximately $5\% - -6\%$ rates implied by the term structure.

Conclusion

The interpolated 3.5-year zero rate is

$$\boxed{z(3.5) \approx 5.6662\%}$$

and the price of the off-benchmark bond is

$$\boxed{P \approx \text{Rs. } 103.60}.$$

Since

$$103.60 > 100,$$

the bond trades at a

$$\boxed{\text{small premium to face value.}}$$

Part D – Fixed and Residual Claims

D1. Insurance Company and Liability Matching

Question:

An insurance company sells a policy today that promises the policyholder a single lump-sum payout of Rs. $20,00,000$ exactly 10 years from now. The company wants to fund this liability today by purchasing a portfolio of government bonds from the C3 market, where only maturities of 1–4 years are available. Explain, using ideas from fixed vs. residual claims, discounting, and the term structure, why the insurer cannot fully "match" this liability using only the available instruments, and describe qualitatively what risk the insurer is left exposed to.

Solution

The insurance company's liability is a **fixed claim** because it promises a predetermined payment of

$$\boxed{\text{Rs. } 20,00,000}$$

at a specific future date, namely 10 years from today.

The insurer therefore needs assets whose future cash flows can reliably cover this liability.

1. Problem with the Available Bonds

The C3 market only provides government bonds with maturities of

$$1,\ 2,\ 3,\ 4 \text{ years.}$$

However, the insurance liability occurs at

$$10 \text{ years.}$$

Therefore, none of the available bonds has a maturity matching the liability.

For example, even if the insurer purchases a 4-year government bond, its principal is received after 4 years, whereas the insurance company needs the money after 10 years.

Thus,

$$\boxed{\text{Available asset maturities: 1–4 years}}$$

but

$$\boxed{\text{Liability maturity: 10 years}}$$

so an exact maturity match is impossible.

2. Role of Discounting

The present value of the 10-year liability depends on the appropriate interest rate for a 10-year maturity.

Conceptually,

$$PV(\text{Liability}) = \frac{\text{Rs. } 20,00,000}{\text{appropriate 10-year discount factor}}.$$

However, the available C3 term structure only gives information about maturities up to 4 years.

Therefore, the insurer does not have a directly available 10-year discount rate from this market.

The insurer can calculate present values using the available 1–4 year rates, but those rates do not completely determine the value of a cash flow occurring 10 years in the future.

3. Role of the Term Structure

The **term structure of interest rates** tells us that interest rates can differ across different maturities.

From C3, the rates generally increase with maturity. However, the curve only extends to 4 years.

To value or hedge the 10-year liability accurately, the insurer would need information about the term structure beyond 4 years, particularly around the 10-year maturity.

Thus, the absence of longer-term government bonds prevents the insurer from constructing a portfolio whose cash flows exactly match the 10-year liability.

4. What Risk Remains?

The insurer is left with **reinvestment risk** and **interest-rate/term-structure risk**.

For example, suppose the insurer invests in a 4-year bond. After 4 years, the bond matures, but the liability is still 6 years away. The insurer must then reinvest the proceeds.

The future interest rates at that time are uncertain.

If interest rates fall, the insurer may earn a lower return on the reinvested money and may not accumulate enough to meet the

$$\text{Rs. } 20,00,000$$

liability after 10 years.

Therefore, the insurer cannot lock in the exact funding required for the 10-year liability using only the 1–4 year bonds.

5. Fixed Claim vs. Residual Claim

The insurance liability is a **fixed claim** because the insurer has a contractual obligation to pay a predetermined amount.

This is different from an equity or residual claim, where the payment depends on the residual performance of the underlying business.

Because the insurer's obligation is fixed, it is particularly important to match the timing and amount of its asset cash flows with the liability.

Here, the available bonds do not provide the required 10-year cash flow.

Final Answer

The insurer cannot fully match the liability because the liability is a fixed payment of

$$\boxed{\text{Rs. } 20,00,000}$$

at year 10, while the available government bonds mature only between years 1 and 4.

Consequently, the insurer cannot construct a portfolio whose cash flows exactly coincide with the timing of the liability, nor can it directly use a 10-year discount rate from the given market.

The insurer therefore remains exposed primarily to **interest-rate and reinvestment risk**. After the short-term bonds mature, the proceeds must be reinvested, and future interest rates may be different from today's rates.

If rates fall, the reinvested funds may grow less than expected, creating a risk that the insurer will not have enough money to meet the 10-year liability.

$$\boxed{\text{No 10-year instrument} \implies \text{No exact liability matching} \implies \text{Reinvestment and interest-rate risk}}$$